

Shashini Marasinghe Mudiyanse

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EDUCATION

Michigan State University, East Lansing, MI

Ph.D. Candidate, Mathematics (expected Summer 2027)

- Thesis topic: Quantum Invariants and the Generalized Turaev–Viro Invariants Volume Conjecture.
- Advisor: Professor Efstratia Kalfagianni.

University of Peradeniya, Sri Lanka

B.Sc. Honours degree in Mathematics, First-Class Honours, 2021.

RESEARCH INTERESTS

Quantum and Low-dimensional Topology, Knot Theory, Quantum Computing, Mathematics Education.

PUBLICATIONS

- Marasinghe, S., “Seifert fibered 3-manifolds and Turaev–Viro invariants volume conjecture,” submitted to the *International Journal of Mathematics*. arXiv:2504.10682.
- Detcherry, R., Kalfagianni, E., Marasinghe, S., “Seifert cobordisms and the Chen–Yang volume conjecture,” submitted to the *Journal of the London Mathematical Society*. arXiv:2505.01546.
- Marasinghe, S., Wong, K.H., Kalfagianni, E., “Gluing of two hyperbolic 3-manifolds and the Generalized Turaev–Viro invariants volume conjecture,” in preparation.

CONFERENCE TALKS

- TBD, IPAM long program, Fall 2026, UCLA.
- *Seifert cobordisms and the Chen–Yang volume conjecture*, Advances in Geometric and Quantum Topology, July 13–16, 2026, Temple University, Philadelphia.
- *Seifert cobordisms and the Chen–Yang volume conjecture*, Pitt AWM Student Seminar, University of Pittsburgh (March 2026).
- *Seifert fibered 3-manifolds and Turaev–Viro invariants volume conjecture*, Midwest Panorama of Geometry and Topology Conference, University of Iowa (June 2025).
- *Seifert fibered 3-manifolds and Turaev–Viro invariants volume conjecture*, 10th Annual Graduate Student Conference in Algebra, Geometry, and Topology, Temple University (May 2025).
- *Witten–Reshetikhin–Turaev invariant of Seifert fibered 3-manifolds*, MSU Graduate Geometry and Topology Seminar (Spring 2025).
- *Introduction to the Volume Conjecture*, MSU Graduate Geometry and Topology Seminar (Spring 2025).
- *6-j symbols*, MSU Graduate Geometry and Topology Seminar (2024).

RESEARCH MENTORING

- RTG MSU Summer Topology Program (Summer 2023): Mentored two undergraduate students on quantum invariants through a knot-theoretic approach, meeting four days a week over five weeks and guiding them

from foundational concepts to a final research presentation. Supervised their mathematical writing and $\text{\LaTeX}$ skills as each student developed a separate section, and coached them in preparing and delivering individual write-ups and presentations.

- Resource person for $\text{\LaTeX}$ workshop in Sri Lanka (2024).

PROFESSIONAL DEVELOPMENT

- Served as the spokesperson for the Volume Conjecture and Beyond working group at the IPAM Long Program (Quantum Topology, Character Varieties, and Low-Dimensional Geometry) (September–December 2026).
- IPAM Long Program (Quantum Topology, Character Varieties and Low-Dimensional Geometry): highly selective research program at the Institute for Pure and Applied Mathematics (UCLA), one of the core participants selected worldwide, focused on quantum topology, character varieties, and low-dimensional geometry (September–December 2026).
- FAST Fellowship (Future Academic Scholars in Teaching): competitive teaching-as-research professional development program for STEM Ph.D. students at Michigan State University, selected as 1 of 9 university-wide. Planned and conducted a mentored classroom research project to improve undergraduate teaching and learning (2025–2026).
- Certification in College Teaching (in progress), MSU Graduate School: completed all coursework, workshop, and mentored teaching requirements, including the two-day Certification in College Teaching Institute (CCTI); final teaching e-portfolio in progress (2025–2026).

HONORS & AWARDS

Graduate Fellowship, Michigan State University, \$15,000	2026–2027
IPAM Long Program Core Participant, Quantum Topology, Character Varieties and Low-Dimensional Geometry, IPAM, UCLA	2026
Herbert T. Graham Scholarship, \$2,000	2025–2026
FAST Fellowship (Future Academic Scholars in Teaching), \$5,000	2025–2026
TA Award for Excellence in Teaching, Michigan State University, \$500	2023–2024
Best Poster Presentation Award, ICCME	2021
Award for Academic Excellence, University of Peradeniya	2021
Dean’s List, University of Peradeniya (two semesters)	2018–2019
Runner-up, EUDOXUS Quiz Competition	2018

SERVICE & LEADERSHIP

Spokesperson, Volume Conjecture and Beyond group, IPAM Long Program (Quantum Topology, Character Varieties, and Low-Dimensional Geometry), IPAM, UCLA	2026
Organizer, MSU Graduate Geometry and Topology Seminar	2024–2025
Treasurer, American Mathematical Society Graduate Student Chapter, MSU	2022–2023
Secretary, Association for Women in Mathematics, MSU Chapter	2021–2022
Lead TA, MSU	2022–2024
Editor, magazine “Mathéma,” University of Peradeniya	2019–2020

TEACHING EXPERIENCE

Instructor of Record (Full responsibility for course design, instruction, and outcomes-aligned assessments.)

MTH 103B – College Algebra B (Online)

Summer 2026

Adapted materials and assessments for synchronous online delivery with remote proctoring. Facilitated daily collaborative group work via breakout rooms and a shared Microsoft Whiteboard for active learning. Used standards-based, four-point mastery grading with reassessment to support student success.

MTH 103A – College Algebra A (In-person)

Spring 2026

Used daily collaborative group work to promote active learning in a 40-student section, with standards-based, four-point mastery grading. Mentored an undergraduate learning assistant supporting in-class activities.

MTH 299 – Introduction to Proofs (Online)

Summer 2025

Facilitated collaborative group work via breakout rooms and a shared Microsoft Whiteboard to promote active learning in a synchronous online section. Integrated $\text{\LaTeX}$ to develop mathematical communication and assessed proof-writing growth through a cumulative final portfolio.

MTH 234 – Multivariable Calculus 3 (In-person)

Fall 2023, Spring 2024

Implemented a modified flipped classroom in a 32-student section, using pre-class videos with graded readiness quizzes to prepare students, and blending instruction with collaborative group work for active learning.

MTH 101 – Quantitative Literacy 1 (Online)

Summer 2023

Taught three fully asynchronous online sections of 100+ students, structuring self-paced modules to sustain engagement, with individualized feedback and extended virtual office hours at scale.

MTH 124 – Survey of Calculus 1 (In-person)

Summer 2022

Used collaborative group activities in a 20-student section for active learning, and adapted assignments to emphasize real-world relevance and applications.

Teaching Team

MTH 299 – Introduction to Proofs (In-person)

Fall 2025

Led two recitation sections of 60 students, using collaborative group work for problem-solving and providing detailed feedback on proof-writing.

MTH 101 – Quantitative Literacy 1 (Online)

Fall 2022

Supported three online sections of 100+ students, facilitating discussions, holding virtual office hours, grading, and resolving WeBWorK and D2L technical issues on the course support team.

MTH 124 – Survey of Calculus 1 (In-person)

Spring 2022

Held office hours and graded assessments, providing individualized feedback to support student learning.

MTH 133 – Calculus 2 (In-person)

Fall 2021

Held problem-solving office hours and provided detailed feedback on student work.

Curriculum Development

MTH 103A – College Algebra A (In-person)

Spring 2026

Designed worksheets and assessments to support cooperative group learning in a high-enrollment gateway course, aimed at increasing student engagement and making the course more approachable.

MTH 299 – Introduction to Proofs (Online)

Summer 2025

Designed proof-writing supports, collaborative group activities, and a $\text{\LaTeX}$ component for a modified flipped online classroom, with a cumulative final portfolio assessing proof-writing growth.

MTH 101 – Quantitative Literacy 1 (Online)

Spring 2023, Spring 2025

Redesigned WeBWorK homework and developed real-world project assignments for a gateway course serving 1000+ students, strengthening quantitative reasoning and engagement.

MTH 234 – Multivariable Calculus 3 (In-person)

Fall 2023, Spring 2024

Developed original worksheets and assessment materials to encourage cooperative learning.

MTH 124 – Survey of Calculus 1 (In-person)

Summer 2022

Redesigned practice materials and guided examples for a small-class setting, supporting interactive, discussion-based instruction.

Teaching Mentoring

- Mentored an undergraduate learning assistant one-on-one in the gateway course MTH 103A (College Algebra A), supporting their facilitation of in-class group activities (2026).
- Instructional Mentoring Lead TA for undergraduate learning assistants: observed 20+ ULAs over a semester and provided structured verbal and written feedback on teaching practices (2022–2024).
- Instructional Mentoring Lead TA for graduate second-year teaching assistants: conducted two classroom observations per GTA and held in-person mentoring meetings with written feedback focused on instructional improvement (2024).
- Resource person for discussion-based instruction for undergraduate learning assistants and first-year teaching assistants (2025).

TEACHING & EDUCATION TALKS

- Patterns Emerge in Students' Use of AI in Mathematics, FAST Fellowship Symposium, Michigan State University (April 2026).

SKILLS

$\text{\LaTeX}$, Python, Lean, Adobe Illustrator, GitHub, D2L, Gradescope, WeBWorK.